

# Exercise 13.2

① If a line makes an angle  $\frac{\pi}{4}$  with each of the  $x$ -axis and  $y$ -axis, what angle does it make with  $z$ -axis.

→ Soln.

The angle made by a line with  $x$ -axis and  $y$ -axis are respectively  $\alpha = \frac{\pi}{4}$  and

$$\beta = \pi/4$$

Now,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\text{or, } \cos^2 \frac{\pi}{4} + \cos^2 \frac{\pi}{4} + \cos^2 \gamma = 1$$

$$\text{or, } \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \cos^2 \gamma = 1$$

$$\text{or, } \frac{1}{2} + \frac{1}{2} + \cos^2 \gamma = 1$$

$$\text{or, } \cos^2 \gamma = 1 - \frac{1}{2} - \frac{1}{2}$$

$$\text{or, } \cos^2 \gamma = 0$$

$$\text{or, } \cos \gamma = \cos 90$$

$$\text{or, } \gamma = 90^\circ = \pi/2$$

∴ The angle made by a line with  $z$ -axis is  $\pi/2$



② Find the direction cosines of a line which is equally inclined with the axes of co-ordinates.

→ Soln,

Given,

$$\alpha = \beta = \gamma = ? \text{ (let be } \eta \text{)}$$

We know that,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\text{or, } \cos^2 \eta + \cos^2 \eta + \cos^2 \eta = 1$$

$$\text{or, } 3 \cos^2 \eta = 1$$

$$\text{or, } \cos^2 \eta = \frac{1}{3}$$

$$\text{or, } \cos \eta = \sqrt{\frac{1}{3}}$$

$$\text{or, } \cos \eta = \pm \frac{1}{\sqrt{3}}$$

Now,

$$\cos \alpha = \pm \frac{1}{\sqrt{3}}, \cos \beta = \pm \frac{1}{\sqrt{3}}, \cos \gamma = \pm \frac{1}{\sqrt{3}}$$

∴ direction cosines are  $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$ .

③ If  $\alpha, \beta$  and  $\gamma$  are the direction angles of a line prove that  $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + 1 = 0$

→ Soln,

$$\text{L.H.S} = \cos 2\alpha + \cos 2\beta + \cos 2\gamma + 1$$

$$= 2\cos^2 \alpha - 1 + 2\cos^2 \beta - 1 + 2\cos^2 \gamma - 1 + 1$$

$$= 2(\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma) - 2$$

$$= 2 \times 1 - 2$$

$$= 2 - 2$$

$$= 0 = \text{R.H.S}$$

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(a) Find the direction cosines of each of the lines whose direction ratios are.

a) 6, 2, -3

→ Soln,

let,  $a=6$ ,  $b=2$  and  $c=-3$

$$\begin{aligned}\text{Now, } \sqrt{a^2+b^2+c^2} &= \sqrt{6^2+2^2+(-3)^2} \\ &= \sqrt{36+4+9} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

So, the direction cosines are:

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{6}{7}$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{2}{7}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{-3}{7}$$

b) -1, -2, -3

→ Soln,

let  $a=-1$ ,  $b=-2$ ,  $c=-3$

$$\begin{aligned}\text{Now, } \sqrt{a^2+b^2+c^2} &= \sqrt{(-1)^2+(-2)^2+(-3)^2} \\ &= \sqrt{1+4+9} \\ &= \sqrt{14}\end{aligned}$$

So, the direction cosines are:

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{-1}{\sqrt{14}}$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{-2}{\sqrt{14}}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{-3}{\sqrt{14}}$$

⑤ Find the direction cosines of each of the following lines joining the points.

a)  $(-2, 1, -8)$  and  $(4, 3, -5)$

→ Soln,

$$\text{let } A(-2, 1, -8) = (x_1, y_1, z_1)$$

$$B(4, 3, -5) = (x_2, y_2, z_2)$$

The direction ratios of AB are,

$$a = x_2 - x_1 = 4 - (-2) = 6$$

$$b = y_2 - y_1 = 3 - 1 = 2$$

$$c = z_2 - z_1 = -5 - (-8) = 3$$

Now,

$$\begin{aligned} \sqrt{a^2+b^2+c^2} &= \sqrt{6^2+2^2+3^2} \\ &= \sqrt{36+4+9} \\ &= \sqrt{49} \\ &= 7 \end{aligned}$$

So, the direction cosines are,

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{6}{7}$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{2}{7}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{3}{7}$$



b) (5, 2, 8) and (7, -1, 9)

→ let  $A(5, 2, 8) = (x_1, y_1, z_1)$

$B(7, -1, 9) = (x_2, y_2, z_2)$

The direction ratios of AB are,

$$a = x_2 - x_1 = 7 - 5 = 2$$

$$b = y_2 - y_1 = -1 - 2 = -3$$

$$c = z_2 - z_1 = 9 - 8 = 1$$

Now,

$$\begin{aligned}\sqrt{a^2 + b^2 + c^2} &= \sqrt{2^2 + (-3)^2 + 1^2} \\ &= \sqrt{4 + 9 + 1} \\ &= \sqrt{14}\end{aligned}$$

So, the direction cosines are,

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}} = \frac{2}{\sqrt{14}}$$

$$m = \frac{b}{\sqrt{a^2 + b^2 + c^2}} = \frac{-3}{\sqrt{14}}$$

$$n = \frac{c}{\sqrt{a^2 + b^2 + c^2}} = \frac{1}{\sqrt{14}}$$

⑥ If the direction cosines of a line PQ are proportional to 2, 4, 4 then find the angles made by PQ with co-ordinates axes.

→ let, a, b and c are directional ratios,  
So,

$$a = 2, b = 4, c = 4$$

Now,

$$\begin{aligned}\sqrt{a^2 + b^2 + c^2} &= \sqrt{2^2 + 4^2 + 4^2} \\ &= \sqrt{4 + 16 + 16} \\ &= \sqrt{36} \\ &= 6\end{aligned}$$



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∴ Then, The directional cosines are,

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{2}{6} = \frac{1}{3}$$

$$\cos \alpha = \frac{1}{3}$$

$$\alpha = \cos^{-1}\left(\frac{1}{3}\right)$$

$$\therefore \alpha = 70.52$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{4}{6} = \frac{2}{3}$$

$$\cos \beta = \frac{2}{3}$$

$$\beta = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\therefore \beta = 48.18$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{4}{6} = \frac{2}{3}$$

$$\cos \gamma = \frac{2}{3}$$

$$\gamma = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\therefore \gamma = 48.18$$

∴ The angle made by PQ with co-ordinates axes are  $70.52^\circ$ ,  $48.18^\circ$  and  $48.18^\circ$  respectively.



7. If  $P(1, 2, 3)$ ,  $Q(3, -6, 7)$  and  $R(2, 1, 1)$  be any three points on the space.

i) Find the mid-point  $M$  of the line joining the point  $P$  and  $Q$ .

→ Soln,

$$\begin{aligned}\text{Midpoint } (x, y, z) &= \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right) \\ &= \left( \frac{1+3}{2}, \frac{2-6}{2}, \frac{3+7}{2} \right) \\ &= \left( \frac{4}{2}, \frac{-4}{2}, \frac{10}{2} \right) \\ &= (2, -2, 5)\end{aligned}$$

∴ Midpoint  $(x, y, z) = (2, -2, 5)$ .

ii) Find the direction ratios of the line  $MR$ .

→ Soln,

$$\begin{aligned}\text{let } M(2, -2, 5) &= (x_1, y_1, z_1) \\ R(2, 1, 1) &= (x_2, y_2, z_2)\end{aligned}$$

Direction Ratios are,

$$a = x_2 - x_1 = 2 - 2 = 0$$

$$b = y_2 - y_1 = 1 - (-2) = 3$$

$$c = z_2 - z_1 = 1 - 5 = -4$$

∴ Direction ratios of the line  $MR$  are  $0, 3, 4$  respectively.



iii) Find the direction cosines of the line MR.

→ Soln,

$$a=0, b=3, c=-4$$

Now,

$$\begin{aligned}\sqrt{a^2+b^2+c^2} &= \sqrt{0^2+3^2+(-4)^2} \\ &= \sqrt{9+16} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

Direction cosines are,

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{0}{5} = 0$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{3}{5}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{-4}{5}$$

∴ Direction cosines of the line MR are  $0, \frac{3}{5}, -\frac{4}{5}$  respectively.

iv) State the angles made by the line MR with positive direction of x-axis, y-axis and z-axis.

→ Soln,

$$l=0$$

$$m=\frac{3}{5}$$

$$\cos \alpha = 0$$

$$\cos \alpha = \cos 90^\circ$$

$$\alpha = 90^\circ$$

$$\cos \beta = 3/5$$

$$\beta = \cos^{-1}(3/5)$$

$$\beta = 53.13^\circ$$



$$n = -4/5$$

$$\cos Y = -4/5$$

$$Y = \cos^{-1}(-4/5)$$

$$Y = 143.13^\circ$$

$$\therefore \alpha = 90^\circ, \beta = 53.13^\circ, \gamma = 143.13^\circ$$

8. Find the direction cosines  $l, m, n$  of two lines which satisfy the equation.

a)  $4l + 3m - 2n = 0$  and  $lm - mn + nl = 0$

→ Soln,

$$4l + 3m - 2n = 0 \rightarrow (i)$$

$$lm - mn + nl = 0 \rightarrow (ii)$$

From eqn (i)

$$2n = 4l + 3m$$

$$n = \frac{4l + 3m}{2} \rightarrow (iii)$$

Putting the value of  $n$  in eqn (ii)

$$lm - m\left(\frac{4l + 3m}{2}\right) + \left(\frac{4l + 3m}{2}\right)l = 0$$

$$\text{or, } lm - \frac{4lm - 3m^2}{2} + \frac{4l^2 + 3lm}{2} = 0$$

$$\text{or, } 2lm - 4lm - 3m^2 + 4l^2 + 3lm = 0$$

$$\text{or, } lm - 3m^2 + 4l^2 = 0$$

$$\text{or, } 3m^2 - lm - 4l^2 = 0$$

$$\text{or, } 3m^2 \text{ comparing eqn with}$$

$$\text{or, } 3m^2 - 4lm + 3lm - 4l^2 = 0$$



$$\text{or, } m(3m-4l) + l(3m-4l) = 0$$

$$\text{or, } (m+l)(3m-4l) = 0$$

∴ either,

$$m+l = 0 \rightarrow \textcircled{\text{iv}} \quad 0 = 3m-4l \rightarrow \textcircled{\text{v}}$$

$$\text{i.e. } m + 0n + l = 0$$

$$\text{i.e. } -4l + 3m + 0n = 0$$

Solving eqn (i) and (iv) by cross multiplication method.

$$\frac{l}{0-2} = \frac{m}{2-0} = \frac{n}{4-3}$$

$$\text{or, } \frac{l}{-2} = \frac{m}{2} = \frac{n}{1}$$

$$\begin{aligned} \text{Now, } \sqrt{a^2+b^2+c^2} &= \sqrt{(-2)^2+2^2+1^2} \\ &= \sqrt{4+4+1} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

Then,

direction cosine are,

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{-2}{3}$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{2}{3}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{1}{3}$$

Similarly,

Solving eqn (i) and (v) by cross multiplication method.

$$\frac{l}{6-0} = \frac{m}{0-(-8)} = \frac{n}{-12-12}$$



or,  $\frac{l}{6} = \frac{m}{8} = \frac{n}{-24}$

$\therefore a=6, b=8, c=-24$

So,

$$\begin{aligned}\sqrt{a^2+b^2+c^2} &= \sqrt{6^2+8^2+(-24)^2} \\ &= \sqrt{36+64+576} \\ &= \sqrt{676} \\ &= 26\end{aligned}$$

Now,

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{6}{26} = \frac{3}{13}$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{8}{26} = \frac{4}{13}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{-24}{26} = \frac{-12}{13}$$

$\therefore$  The direction cosines  $l, m, n$  of two lines are  $-\frac{2}{3}, \frac{2}{3}, \frac{1}{3}$  and  $\frac{3}{13}, \frac{4}{13}, \frac{-12}{13}$

respectively.

b)  $2l+2m-n=0$  and  $mn+nl+lm=0$

$\rightarrow$  Soln,

$$2l+2m-n=0 \rightarrow (i)$$

$$mn+nl+lm=0 \rightarrow (ii)$$

From eqn (i)

$$n=2l+2m \rightarrow (iii)$$

putting the value of  $n$  in eqn (ii)



$$m(2l+2m) + (2l+2m)l + lm = 0$$

$$\text{or, } 2lm + 2m^2 + 2l^2 + 2ml + lm = 0$$

$$\text{or, } 2m^2 + 5lm + 2l^2 = 0$$

$$\text{or, } 2m^2 + 4lm + lm + 2l^2 = 0$$

$$\text{or, } 2m(m+2l) + l(m+2l) = 0$$

$$\text{or, } (2m+l)(m+2l) = 0$$

Either,

$$2m+l=0 \rightarrow \text{(iv)}$$

$$m+2l=0$$

$$\text{i.e. } l+2m+0n=0 \rightarrow \text{i.e. } 2l+m+0n=0$$

Solving eqn (i) and (iv) by cross multiplication method.

$$\frac{l}{0-(-2)} = \frac{m}{-1-0} = \frac{n}{4-2}$$

$$\text{or, } \frac{l}{+2} = \frac{m}{-1} = \frac{n}{2}$$

Now,

$$\begin{aligned} a=2, b=-1, c=2 \\ \sqrt{a^2+b^2+c^2} &= \sqrt{2^2+(-1)^2+2^2} \\ &= \sqrt{4+1+4} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

Then,

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{2}{3}$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}} = \frac{-1}{3}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}} = \frac{2}{3}$$



Similarly,

Solving eqn (i) and (v) by cross multiplication method.

$$\frac{L}{0 - (-1)} = \frac{m}{-2 - 0} = \frac{n}{2 - 4}$$

or,  $\frac{L}{1} = \frac{m}{-2} = \frac{n}{-2}$

So,

$$a = 1, b = -2 \text{ and } c = -2$$

Now,

$$\begin{aligned}\sqrt{a^2 + b^2 + c^2} &= \sqrt{1^2 + (-2)^2 + (-2)^2} \\ &= \sqrt{1 + 4 + 4} \\ &= \sqrt{9} \\ &= 3\end{aligned}$$

Again,

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}} = \frac{1}{3}$$

$$m = \frac{b}{\sqrt{a^2 + b^2 + c^2}} = \frac{-2}{3}$$

$$n = \frac{c}{\sqrt{a^2 + b^2 + c^2}} = \frac{-2}{3}$$

$\therefore$  The direction cosines  $l, m, n$  of two lines are  $\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}$  and  $\frac{1}{3}, -\frac{2}{3}$  and  $-\frac{2}{3}$

respectively.